

Measurement of Thermal Energy

Form Three Revision Notes

1 Temperature, Thermal Energy and the Kinetic Picture

Every material is made of tiny particles (atoms or molecules) that are always moving. In a solid they vibrate about fixed positions; in a liquid they slide past one another; in a gas they fly about freely. Because they move, the particles carry *kinetic energy*, and because they attract one another they also store *potential energy*. The total of all this microscopic energy is the internal (thermal) energy of the body.

Definition

Temperature is a measure of the average kinetic energy of the particles of a body. It tells us how hot or cold a body is, but not how much heat energy the body contains. Its SI unit is the kelvin (K); everyday work often uses degrees Celsius ($^{\circ}\text{C}$).

Definition

Thermal (internal) energy is the total kinetic and potential energy stored in all the particles of a body. It is a form of energy and is measured in joules (J).

NB

Temperature and heat are not the same thing. A cup of boiling water is hotter than a warm bath, yet the bath contains far more heat energy because it has a much greater mass. Convert temperatures with $T/\text{K} = \theta/^{\circ}\text{C} + 273$. A *change* of temperature has the same size in K and $^{\circ}\text{C}$, so $\Delta\theta$ may be quoted in either unit.

NB

Symbols used in the worked solutions. Q = heat energy (J); m = mass (kg); c = specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$); C = heat capacity (J K^{-1}); $\Delta\theta$ = temperature change (K); θ = temperature ($^{\circ}\text{C}$); L_f = specific latent heat of fusion, L_v = specific latent heat of vapourisation (both J kg^{-1}); P = power (W); t = time (s). Each solution below sets out the **Given** data, the **Required** unknown, the **Formula**, the step-by-step **Working** and the final **Answer**.

The quantity of heat needed to warm a body depends on three things:

- the **mass** of the body (more mass needs more heat),
- the **temperature change** required (a bigger rise needs more heat),
- the **nature of the material** (different substances soak up heat differently).

3 Heat Capacity

Definition

Heat capacity C of a body is the amount of heat energy needed to raise the temperature of the *whole body* by one kelvin (or one degree Celsius):

$$C = \frac{Q}{\Delta\theta}, \quad \text{unit: J K}^{-1}.$$

NB

Heat capacity refers to a particular object, so a large kettle of water has a bigger heat capacity than a small cup of the same water. It does not tell us anything about the material on its own.

Worked Example : heat capacity of a block

A heater delivers 9600 J of energy to an iron block and its temperature rises from 20°C to 60°C . Find the heat capacity of the block.

Given: heat supplied $Q = 9600 \text{ J}$; initial temperature $\theta_1 = 20^{\circ}\text{C}$; final temperature $\theta_2 = 60^{\circ}\text{C}$.

Required: the heat capacity C (J K^{-1}).

Formula: $C = \frac{Q}{\Delta\theta}$, where $\Delta\theta = \theta_2 - \theta_1$.

Working:

$$\Delta\theta = 60 - 20 = 40 \text{ K}$$

$$C = \frac{Q}{\Delta\theta} = \frac{9600 \text{ J}}{40 \text{ K}} = 240 \text{ J K}^{-1}.$$

Answer: $C = 240 \text{ J K}^{-1}$

2 Heat and What Determines It

When two bodies at different temperatures are placed together, energy flows from the hotter to the cooler one until they reach the same temperature. That energy in transit is called heat.

Definition

Heat is the energy transferred from one body to another because of a difference in temperature. Like all energy it is measured in joules (J).

4 Specific Heat Capacity

Definition

Specific heat capacity c of a substance is the amount of heat energy needed to raise the temperature of 1 kg of the substance by one kelvin:

$$c = \frac{Q}{m \Delta\theta}, \quad \text{unit: J kg}^{-1} \text{K}^{-1}.$$

Rearranging gives the heat equation used constantly in

this topic:

$$Q = mc\Delta\theta$$

NB

Water has an unusually large specific heat capacity ($4200 \text{ J kg}^{-1} \text{ K}^{-1}$). This is why water heats up slowly, cools down slowly and is used as a coolant in car engines and as a store of heat in hot-water systems.

Typical specific heat capacities

Substance	$c \text{ (J kg}^{-1} \text{ K}^{-1}\text{)}$
Water	4200
Ice	2100
Steam	2010
Aluminium	900
Iron	480
Copper	390
Brass	320
Lead	130

Worked Example : warming water

How much heat is needed to raise the temperature of 2.5 kg of water from 20°C to 80°C ? Take $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

Given: mass $m = 2.5 \text{ kg}$; specific heat capacity $c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$; $\theta_1 = 20^\circ\text{C}$, $\theta_2 = 80^\circ\text{C}$.

Required: the heat supplied $Q \text{ (J)}$.

Formula: $Q = mc\Delta\theta$, with $\Delta\theta = \theta_2 - \theta_1$.

Working:

$$\begin{aligned}\Delta\theta &= 80 - 20 = 60 \text{ K} \\ Q &= mc\Delta\theta \\ &= 2.5 \times 4200 \times 60 \\ &= 630\,000 \text{ J} = 630 \text{ kJ}.\end{aligned}$$

Answer: $Q = 630 \text{ kJ}$

Worked Example : heating a metal block

An aluminium block of mass 0.8 kg is heated from 25°C to 175°C . Find the heat supplied. Take $c_{\text{Al}} = 900 \text{ J kg}^{-1} \text{ K}^{-1}$.

Given: mass $m = 0.8 \text{ kg}$; $c = 900 \text{ J kg}^{-1} \text{ K}^{-1}$; $\theta_1 = 25^\circ\text{C}$, $\theta_2 = 175^\circ\text{C}$.

Required: the heat supplied $Q \text{ (J)}$.

Formula: $Q = mc\Delta\theta$, with $\Delta\theta = \theta_2 - \theta_1$.

Working:

$$\begin{aligned}\Delta\theta &= 175 - 25 = 150 \text{ K} \\ Q &= mc\Delta\theta = 0.8 \times 900 \times 150 \\ &= 108\,000 \text{ J} = 108 \text{ kJ}.\end{aligned}$$

Answer: $Q = 108 \text{ kJ}$

Worked Example : finding specific heat capacity

A 0.5 kg copper cylinder is given 7800 J of heat and its temperature rises by 40 K. Find its specific heat capacity.

Given: mass $m = 0.5 \text{ kg}$; heat supplied $Q = 7800 \text{ J}$; temperature change $\Delta\theta = 40 \text{ K}$.

Required: the specific heat capacity $c \text{ (J kg}^{-1} \text{ K}^{-1}\text{)}$.

Formula: $Q = mc\Delta\theta \Rightarrow c = \frac{Q}{m\Delta\theta}$.

Working:

$$\begin{aligned}c &= \frac{Q}{m\Delta\theta} = \frac{7800}{0.5 \times 40} \\ &= \frac{7800}{20} = 390 \text{ J kg}^{-1} \text{ K}^{-1}.\end{aligned}$$

This matches the value for copper in the table above.

Answer: $c = 390 \text{ J kg}^{-1} \text{ K}^{-1}$

5 Calorimetry and the Method of Mixtures

A **calorimeter** is a well-lagged metal container used to measure quantities of heat. When a hot body is placed with a cold one inside it, they exchange heat until they settle at a common (*equilibrium*) temperature.

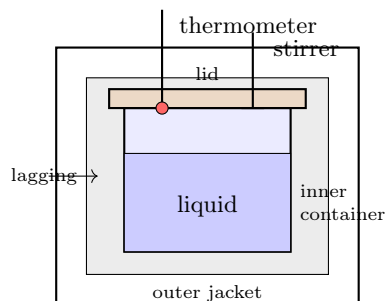


Fig. 1 A lagged calorimeter used in mixtures experiments. (See the corresponding figure in your textbook's chapter on thermal energy.)

Definition

The principle of mixtures (conservation of energy): if no heat escapes to the surroundings, the heat lost by the hot body equals the heat gained by the cold body:

$$Q_{\text{lost}} = Q_{\text{gained}}$$

NB

The container (calorimeter) and stirrer also absorb heat. In careful work their heat capacity is included. In these notes we assume a perfectly lagged calorimeter of negligible heat capacity unless stated otherwise.

Worked Example : equilibrium temperature

0.3 kg of water at 80°C is mixed with 0.5 kg of water at 20°C . Find the final temperature θ .

Given: hot water $m_1 = 0.3 \text{ kg}$ at $\theta_1 = 80^\circ\text{C}$; cold water $m_2 = 0.5 \text{ kg}$ at $\theta_2 = 20^\circ\text{C}$; same liquid, so c is common.

Required: the equilibrium temperature θ ($^\circ\text{C}$).

Formula: principle of mixtures: $Q_{\text{lost}} = Q_{\text{gained}}$, i.e. $m_1 c(\theta_1 - \theta) = m_2 c(\theta - \theta_2)$ (c cancels).

Working:

$$m_1 c(80 - \theta) = m_2 c(\theta - 20)$$

$$0.3(80 - \theta) = 0.5(\theta - 20)$$

$$24 - 0.3\theta = 0.5\theta - 10$$

$$34 = 0.8\theta$$

$$\theta = 42.5^\circ\text{C}.$$

Answer: $\theta = 42.5^\circ\text{C}$

Worked Example : specific heat capacity of a metal

A 0.4 kg metal block heated to 100°C is dropped into 0.2 kg of water at 18°C in a calorimeter of negligible heat capacity. The mixture settles at 30°C . Find the specific heat capacity c_m of the metal.

Given: metal $m_m = 0.4 \text{ kg}$ at 100°C ; water $m_w = 0.2 \text{ kg}$, $c_w = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ at 18°C ; equilibrium $\theta = 30^\circ\text{C}$.

Required: the specific heat capacity c_m of the metal.

Formula: $Q_{\text{lost by metal}} = Q_{\text{gained by water}}$, i.e. $m_m c_m(100 - \theta) = m_w c_w(\theta - 18)$.

Working:

$$m_m c_m(100 - 30) = m_w c_w(30 - 18)$$

$$0.4 c_m \times 70 = 0.2 \times 4200 \times 12$$

$$28 c_m = 10080$$

$$c_m = 360 \text{ J kg}^{-1} \text{ K}^{-1}.$$

Answer: $c_m = 360 \text{ J kg}^{-1} \text{ K}^{-1}$

Worked Example : finding an unknown mass

An unknown mass m of water at 15°C is added to 0.6 kg of water at 75°C . The mixture reaches 55°C . Find m (ignore the container).

Given: cold water mass m (unknown) at 15°C ; hot water $m_2 = 0.6 \text{ kg}$ at 75°C ; equilibrium $\theta = 55^\circ\text{C}$; common c (water).

Required: the unknown mass m (kg).

Formula: $Q_{\text{gained by cold}} = Q_{\text{lost by hot}}$, i.e. $m c(\theta - 15) = m_2 c(75 - \theta)$ (c cancels).

Working:

$$m c(55 - 15) = 0.6 c(75 - 55)$$

$$40 m = 0.6 \times 20$$

$$40 m = 12$$

$$m = 0.3 \text{ kg}.$$

Answer: $m = 0.3 \text{ kg}$

Worked Example : specific heat capacity of a liquid

A heater rated 50 W warms 0.25 kg of an oil by 16 K in 2 min . Assuming no losses, find the specific heat capacity of the oil.

Given: power $P = 50 \text{ W}$; mass $m = 0.25 \text{ kg}$; temperature change $\Delta\theta = 16 \text{ K}$; time $t = 2 \text{ min} = 120 \text{ s}$.

Required: the specific heat capacity c of the oil.

Formula: heat delivered $Q = P t$, then $c = \frac{Q}{m \Delta\theta}$.

Working:

$$Q = P t = 50 \times (2 \times 60) = 6000 \text{ J}$$

$$c = \frac{Q}{m \Delta\theta} = \frac{6000}{0.25 \times 16}$$

$$= \frac{6000}{4} = 1500 \text{ J kg}^{-1} \text{ K}^{-1}.$$

Answer: $c = 1500 \text{ J kg}^{-1} \text{ K}^{-1}$

6 Change of State and the Kinetic Theory

Matter exists in three common states: solid, liquid and gas. Adding or removing heat can change one state into another. The kinetic theory explains this by how strongly the particles are held together and how fast they move.

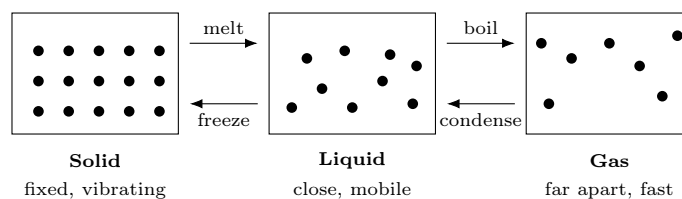


Fig. 2 The three states of water and the changes between them. (See the corresponding figure in your textbook.)

Definition

Melting is the change from solid to liquid at a fixed temperature called the **melting point**. **Freezing** is the reverse change from liquid to solid at the same temperature (the **freezing point**). For a pure substance the two are equal.

NB

While a pure solid melts, the temperature stays constant even though heat is still being supplied. The energy goes into breaking the bonds between particles, not into raising their speed.

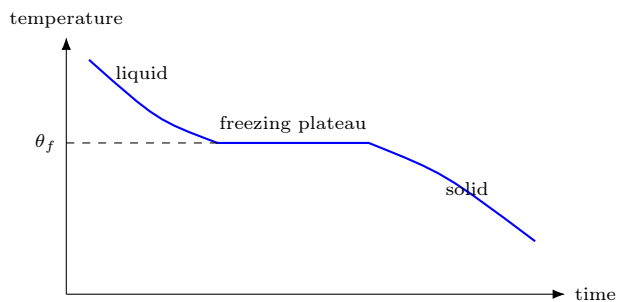


Fig. 3 Cooling curve: a liquid cooling to a solid shows a flat plateau at the freezing point θ_f . (See the corresponding figure in your textbook.)

Worked Example : cooling through the freezing point

A pure substance of mass 0.2 kg is liquid at 90°C and freezes at 60°C . Its specific heat capacity as a liquid is $2500\text{ J kg}^{-1}\text{ K}^{-1}$ and its specific latent heat of fusion is $80\,000\text{ J kg}^{-1}$. Find the total heat removed to cool the liquid to 60°C and then freeze it completely.

Given: mass $m = 0.2\text{ kg}$; liquid at 90°C ; freezing point 60°C ; $c = 2500\text{ J kg}^{-1}\text{ K}^{-1}$; $L_f = 80\,000\text{ J kg}^{-1}$.

Required: the total heat removed Q_{total} (J).

Formula: cool the liquid with $Q_1 = mc\Delta\theta$, then freeze it with $Q_2 = mL_f$; total $Q_{\text{total}} = Q_1 + Q_2$.

Working:

$$Q_1 = mc\Delta\theta = 0.2 \times 2500 \times (90 - 60) = 15\,000\text{ J}$$

$$Q_2 = mL_f = 0.2 \times 80\,000 = 16\,000\text{ J}$$

$$Q_{\text{total}} = Q_1 + Q_2 = 31\,000\text{ J} = 31\text{ kJ}.$$

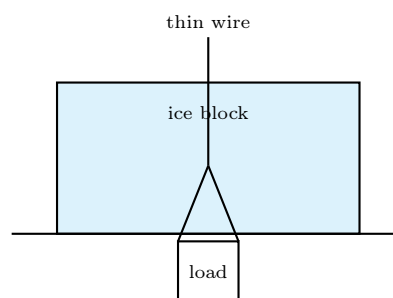
Answer: $Q_{\text{total}} = 31\text{ kJ}$

7 Effect of Impurities and Pressure; Regelation

- **Impurities** lower the melting/freezing point. Salt spread on icy roads makes the ice melt below 0°C .
- **Increasing pressure lowers** the melting point of ice (ice is unusual because it expands on freezing). For most substances higher pressure raises the melting point.

Definition

Regelation is the melting of ice under pressure and its re-freezing once the pressure is removed. A thin wire loaded with weights slowly passes through a block of ice while the block stays in one piece behind it.



wire melts ice below it; water refreezes above

Fig. 4 Regelation: a loaded wire cuts through ice which re-freezes behind it. (See the corresponding figure in your textbook.)

8 Boiling and Evaporation

Both boiling and evaporation turn a liquid into a gas, but they are not the same process.

Definition

Boiling is the rapid change of a liquid to vapour throughout the whole liquid at a fixed temperature, the **boiling point**. **Evaporation** is the slow escape of the fastest molecules from the surface of a liquid, and it happens at all temperatures.

Boiling	Evaporation
Occurs at a fixed temperature	Occurs at any temperature
Throughout the liquid	Only at the surface
Bubbles form in the bulk	No bubbles
Needs continuous heating	Can occur without added heat

The boiling point is affected by:

- **Impurities:** dissolved substances raise the boiling point (salty water boils above 100°C).
- **Pressure:** higher pressure raises the boiling point (a pressure cooker cooks faster). Lower pressure lowers it.
- **Altitude:** on high mountains the air pressure is low, so water boils below 100°C .

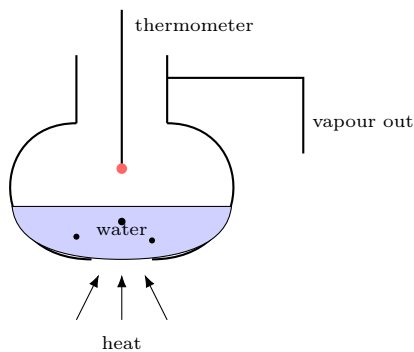


Fig. 5 Boiling apparatus for observing the boiling point. (See the corresponding figure in your textbook.)

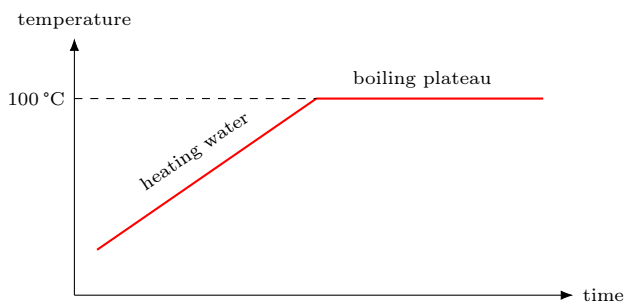


Fig. 6 Heating water: the temperature rises then stays flat while the water boils. (See the corresponding figure in your textbook.)

9 Latent Heat

During a change of state heat is absorbed or released with *no change in temperature*. This “hidden” heat is called latent heat.

Definition

Specific latent heat of fusion L_f is the heat needed to change 1 kg of a solid into liquid at its melting point without a change in temperature. **Specific latent heat of vapourisation** L_v is the heat needed to change 1 kg of a liquid into vapour at its boiling point. In both cases:

$$Q = mL, \quad \text{unit of } L : \text{J kg}^{-1}.$$

NB

For water: $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$ and $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$. Vapourisation absorbs far more energy than fusion, which is why steam at 100°C causes worse scalds than the same mass of boiling water.

Worked Example : melting ice

Find the heat needed to melt 0.4 kg of ice already at 0°C . Take $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$.

Given: mass $m = 0.4 \text{ kg}$ of ice at 0°C ; $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$.

Required: the heat Q needed to melt it (J).

Formula: at the melting point (no temperature change)
 $Q = mL_f$.

Working:

$$\begin{aligned} Q &= mL_f = 0.4 \times 3.34 \times 10^5 \\ &= 133\,600 \text{ J} \approx 134 \text{ kJ}. \end{aligned}$$

Answer: $Q \approx 134 \text{ kJ}$

Worked Example : ice to steam in stages

Find the total heat to change 0.1 kg of ice at -10°C into steam at 120°C . Use $c_{\text{ice}} = 2100$, $c_{\text{water}} = 4200$, $c_{\text{steam}} = 2010 \text{ (J kg}^{-1} \text{ K}^{-1})$, $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$, $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$.

Given: mass $m = 0.1 \text{ kg}$; $c_{\text{ice}} = 2100$, $c_{\text{water}} = 4200$, $c_{\text{steam}} = 2010 \text{ (J kg}^{-1} \text{ K}^{-1})$; $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$, $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$; ice at -10°C to steam at 120°C .

Required: the total heat Q_{tot} (J).

Formula: five stages: warm ice ($Q_1 = mc_{\text{ice}}\Delta\theta$), melt ($Q_2 = mL_f$), warm water ($Q_3 = mc_w\Delta\theta$), boil ($Q_4 = mL_v$), warm steam ($Q_5 = mc_{\text{st}}\Delta\theta$); $Q_{\text{tot}} = Q_1 + Q_2 + Q_3 + Q_4 + Q_5$.

Working:

$$\begin{aligned} Q_1 &= mc_{\text{ice}} \Delta\theta = 0.1 \times 2100 \times 10 = 2100 \text{ J} \\ Q_2 &= mL_f = 0.1 \times 3.34 \times 10^5 = 33\,400 \text{ J} \\ Q_3 &= mc_w \Delta\theta = 0.1 \times 4200 \times 100 = 42\,000 \text{ J} \\ Q_4 &= mL_v = 0.1 \times 2.26 \times 10^6 = 226\,000 \text{ J} \\ Q_5 &= mc_{\text{st}} \Delta\theta = 0.1 \times 2010 \times 20 = 4020 \text{ J} \\ Q_{\text{tot}} &= Q_1 + Q_2 + Q_3 + Q_4 + Q_5 \\ &= 307\,520 \text{ J} \approx 308 \text{ kJ}. \end{aligned}$$

Answer: $Q_{\text{tot}} \approx 308 \text{ kJ}$

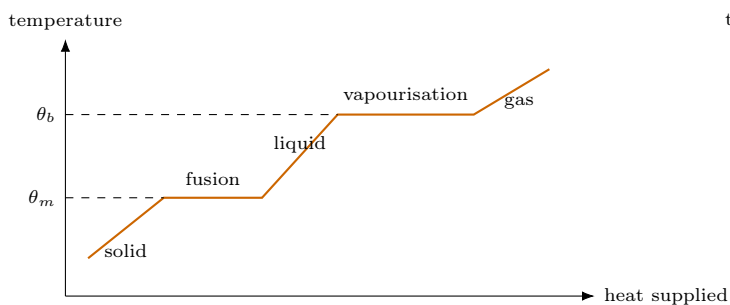


Fig. 7 Temperature versus heat supplied: two flat plateaus mark fusion and vapourisation. (See the corresponding figure in your textbook.)

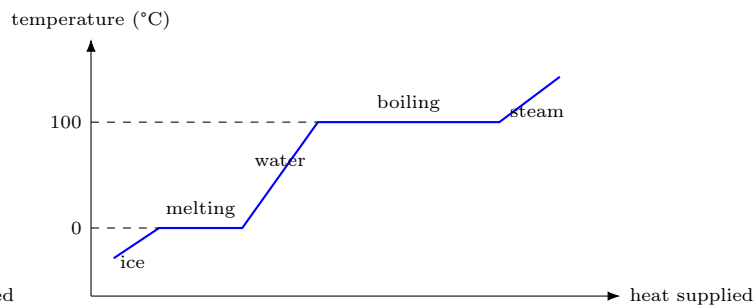


Fig. 8 Staged heating of ice at -10°C to steam at 120°C . (See the corresponding figure in your textbook.)

Worked Example : steam condensing to warm water

Steam at 100°C is passed into 0.5 kg of water at 20°C until the water reaches 40°C . The condensed steam then cools from 100°C to 40°C . Find the mass m_s of steam condensed. Use $L_v = 2.26 \times 10^6\text{ J kg}^{-1}$, $c_w = 4200$.

Given: cold water $m_w = 0.5\text{ kg}$ warmed from 20 to 40°C ; steam mass m_s (unknown) condensing at 100°C then cooling to 40°C ; $L_v = 2.26 \times 10^6\text{ J kg}^{-1}$, $c_w = 4200\text{ J kg}^{-1}\text{ K}^{-1}$.

Required: the mass of steam condensed m_s (kg).

Formula: heat released by steam = heat gained by cold water: $m_s L_v + m_s c_w(100 - 40) = m_w c_w(40 - 20)$.

Working:

$$\begin{aligned} m_s L_v + m_s c_w(100 - 40) &= m_w c_w(40 - 20) \\ m_s(2.26 \times 10^6 + 4200 \times 60) &= 0.5 \times 4200 \times 20 \\ m_s(2.26 \times 10^6 + 2.52 \times 10^5) &= 42000 \\ m_s \times 2.512 \times 10^6 &= 42000 \\ m_s &\approx 0.0167\text{ kg} = 16.7\text{ g}. \end{aligned}$$

Answer: $m_s \approx 16.7\text{ g}$

Worked Example : energy and time to melt lead

A heater rated 300 W melts 0.5 kg of lead already at its melting point. Take $L_f(\text{lead}) = 25\,000\text{ J kg}^{-1}$. Find the energy required and the time taken, assuming no heat is lost.

Given: power $P = 300\text{ W}$; mass $m = 0.5\text{ kg}$ at the melting point; $L_f = 25\,000\text{ J kg}^{-1}$.

Required: the energy Q (J) and the time t (s).

Formula: $Q = mL_f$, then $t = \frac{Q}{P}$.

Working:

$$\begin{aligned} Q &= mL_f = 0.5 \times 25000 = 12\,500\text{ J} \\ t &= \frac{Q}{P} = \frac{12500}{300} \approx 41.7\text{ s}. \end{aligned}$$

Answer: $Q = 12.5\text{ kJ}$, $t \approx 41.7\text{ s}$

fridge, giving that heat out to the room.

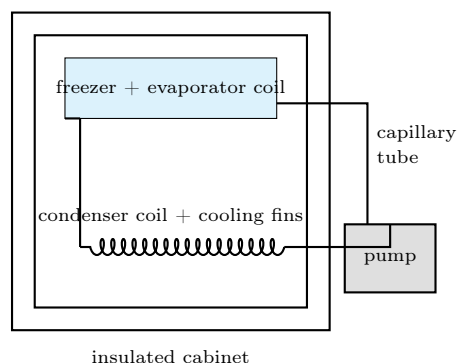


Fig. 9 Domestic refrigerator: evaporator coil in the freezer, capillary tube, compressor pump and condenser fins. (See the corresponding figure in your textbook.)

NB

The pump keeps the refrigerant circulating. The refrigerant *evaporates* in the cold part (absorbing heat) and *condenses* in the warm part (releasing heat), so heat is pumped from inside the fridge to the room.

11 Graph-Reading Practice

Fig. 10 shows a solid being heated steadily until it becomes a gas. Regions AB and CD are sloping (temperature rising) while BC and DE are flat (change of state). Reading such a graph: sloping parts use $Q = mc\Delta\theta$; flat plateaus use $Q = mL$.

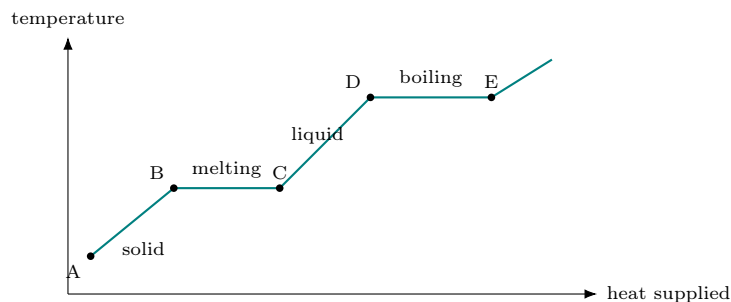


Fig. 10 Generic heating graph for a solid taken through melting and boiling. (See the corresponding figure in your textbook.)

10 Cooling by Evaporation and the Refrigerator

When a liquid evaporates, the fastest molecules escape, so the average kinetic energy of those left behind falls. This is why evaporation cools: sweat drying on the skin, or a wet cloth round a water pot, both feel cold. Volatile liquids (methylated spirit, ether) evaporate fast and produce strong cooling.

Definition

A **refrigerator** uses a volatile working fluid (refrigerant) that evaporates inside the freezer coils, taking latent heat from the food, and then condenses in the coils behind the

Chapter Summary

- **Temperature** measures the average kinetic energy of particles; **heat** is energy transferred because of a temperature difference. Both differ in meaning.
- Heat supplied depends on mass, temperature change and the nature of the material.
- **Heat capacity** $C = Q/\Delta\theta$ (J K^{-1}) applies to a whole object.
- **Specific heat capacity** $c = Q/(m\Delta\theta)$; use $Q = mc\Delta\theta$. Water's large c ($4200\text{ J kg}^{-1}\text{ K}^{-1}$) makes it a good coolant and heat store.
- **Method of mixtures:** in a lagged calorimeter, heat lost = heat gained.
- Matter has three states; the **kinetic theory** ex-

plains state changes by particle spacing and speed.

- **Melting/boiling** happen at fixed temperatures; the temperature stays constant during a change of state.
- **Latent heat of fusion** L_f (solid→liquid) and **vapourisation** L_v (liquid→gas): $Q = mL$; for water $L_v \gg L_f$.
- **Impurities** lower the melting point but raise the boiling point; **pressure** lowers ice's melting point (regelation) and raises the boiling point.
- **Evaporation cools**; a refrigerator uses evaporation and condensation of a refrigerant to pump heat out of the cabinet.

Part C — Method of mixtures

- Q13.** 0.4 kg of water at 70 °C is mixed with 0.6 kg of water at 20 °C. Find the final temperature (ignore the container).
- Q14.** A 0.5 kg copper block ($c = 390$) heated to 95 °C is dropped into 0.3 kg of water at 22 °C in a calorimeter of negligible heat capacity. Find the final temperature.
- Q15.** An aluminium cup of mass 0.15 kg ($c = 900$) holds 0.25 kg of water, all at 18 °C. A 0.2 kg iron block ($c = 480$) at 150 °C is lowered in. Find the common temperature reached (assume no heat is lost).

Part D — Latent heat

- Q16.** Find the heat needed to melt 0.25 kg of ice that is already at 0 °C.
- Q17.** Find the heat needed to turn 0.2 kg of ice at 0 °C into water at 30 °C.
- Q18.** How much heat must be removed to freeze 0.5 kg of water already at 0 °C?
- Q19.** A 500 W immersion heater is switched on for 2 min in a mixture of ice and water at 0 °C. What mass of ice melts (no losses)?
- Q20.** Find the heat needed to change 0.3 kg of water at 50 °C into steam at 100 °C.
- Q21.** A refrigerator freezes 0.6 kg of water already at 0 °C into ice at 0 °C in 4 min. Find the average rate at which it removes heat.

Part E — Graph reading

- Q22.** 0.5 kg of a pure solid is heated steadily and gives the graph of Fig. 10. From the graph, the sloping part AB (solid warming) runs from 0 °C to 50 °C and absorbs 21 000 J; the flat part BC is at 50 °C and the flat part DE is at 120 °C; the boiling plateau DE absorbs 565 000 J.
- Write down the melting point and the boiling point.
 - Find the specific heat capacity of the solid.
 - Find the specific latent heat of vapourisation.

12 Practice Questions

Work through these on your own, then check your method against the **Solutions** section at the end of the notes. Take $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{steam}} = 2010 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J kg}^{-1}$, $L_v(\text{water}) = 2.26 \times 10^6 \text{ J kg}^{-1}$, and other c values from the table on page 1.

Part A — Conceptual / short answer

- Q1.** Explain, using the kinetic theory, why standing in a breeze with wet skin makes you feel cold.
- Q2.** Give two reasons why water is chosen as the coolant in a car engine.
- Q3.** State three differences between boiling and evaporation.
- Q4.** List three factors that change the boiling point of a liquid, and say in which direction each one moves it.
- Q5.** Describe three ways of speeding up the evaporation of a wet surface.
- Q6.** Explain why a drink is cooled more effectively by a lump of ice at 0 °C than by the same mass of water at 0 °C.
- Q7.** A loaded wire slowly cuts through a block of ice, yet the block stays whole. Name this effect and explain it.
- Q8.** Outline how a refrigerator uses a volatile fluid to move heat from the food inside it to the room.
- Q9.** State the effect of dissolving salt in water on (i) its freezing point and (ii) its boiling point.

Part B — Calculations using Q

- Q10.** How much heat is needed to raise the temperature of 3 kg of water from 15 °C to 65 °C?
- Q11.** An aluminium saucepan of mass 1.5 kg is heated from 20 °C to 120 °C. Find the heat absorbed. ($c_{\text{Al}} = 900$.)
- Q12.** The same 12 000 J of heat is given separately to 0.5 kg of water and to 0.5 kg of iron ($c_{\text{iron}} = 480$). Find the temperature rise of each and comment on the difference.

Solutions to Practice Questions

Standard constants used throughout: $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{steam}} = 2010 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{Al}} = 900 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{iron}} = 480 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{\text{copper}} = 390 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J kg}^{-1}$, $L_v(\text{water}) = 2.26 \times 10^6 \text{ J kg}^{-1}$.

Part A — Conceptual

Q1. Water on the skin evaporates, and it is the fastest (most energetic) molecules that escape. This lowers the average kinetic energy of the water and skin left behind, so their temperature falls. A breeze sweeps the vapour away, keeping the escape going, so the cooling is faster and you feel colder.

Q2. (i) Water has a very large specific heat capacity, so it can carry away a lot of heat for only a small rise in its own temperature. (ii) It is cheap, plentiful and easy to pump around the engine. (Its high latent heat and non-flammability are also acceptable reasons.)

Q3. Boiling occurs at one fixed temperature, throughout the whole liquid, with bubbles forming in the bulk and needs a steady supply of heat. Evaporation occurs at any temperature, only at the surface, forms no bubbles and can happen with no added heat.

Q4. (i) Dissolved impurities *raise* the boiling point. (ii) Increased external pressure *raises* the boiling point (lower pressure lowers it). (iii) Higher altitude means lower pressure, which *lowers* the boiling point.

Q5. Increase the surface area (spread the liquid out), raise the temperature, and blow a draught of dry air across it to carry the vapour away. (Reducing the surrounding pressure or humidity also works.)

Q6. Both are at 0°C , but the ice must first absorb its latent heat of fusion ($3.34 \times 10^5 \text{ J kg}^{-1}$) from the drink in order to melt, before the melt-water warms up. The water at 0°C takes only the much smaller $Q = mc\Delta\theta$ as it warms. So the ice removes far more heat from the drink and cools it more.

Q7. This is **regelation**. The pressure of the wire lowers the melting point of the ice directly beneath it, so it melts; the melt-water flows above the wire where the pressure is normal, and re-freezes, giving out latent heat. The wire moves down while the block stays in one piece.

Q8. A pump circulates a volatile refrigerant. In the evaporator coils inside the cabinet the refrigerant evaporates, absorbing its latent heat from the food and cooling it. The vapour is pumped to the condenser coils behind the fridge, where it is compressed and condenses, releasing that latent heat to the room. The cycle repeats, pumping heat from inside to outside.

Q9. Adding salt (i) *lowers* the freezing point of the water and (ii) *raises* its boiling point.

Part B — $Q = mc\Delta\theta$

Q10.

Given: $m = 3 \text{ kg}$ water, $c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$, $\theta_1 = 15^\circ\text{C}$, $\theta_2 = 65^\circ\text{C}$.

Required: the heat supplied Q .

Formula: $Q = mc\Delta\theta$, $\Delta\theta = \theta_2 - \theta_1$.

Working:

$$\begin{aligned}\Delta\theta &= 65 - 15 = 50 \text{ K} \\ Q &= mc\Delta\theta = 3 \times 4200 \times 50 \\ &= 630\,000 \text{ J} = 630 \text{ kJ}.\end{aligned}$$

Answer: $Q = 630 \text{ kJ}$

Q11.

Given: $m = 1.5 \text{ kg}$ aluminium, $c_{\text{Al}} = 900 \text{ J kg}^{-1} \text{ K}^{-1}$, $\theta_1 = 20^\circ\text{C}$, $\theta_2 = 120^\circ\text{C}$.

Required: the heat absorbed Q .

Formula: $Q = mc\Delta\theta$, $\Delta\theta = \theta_2 - \theta_1$.

Working:

$$\begin{aligned}\Delta\theta &= 120 - 20 = 100 \text{ K} \\ Q &= mc\Delta\theta = 1.5 \times 900 \times 100 \\ &= 135\,000 \text{ J} = 135 \text{ kJ}.\end{aligned}$$

Answer: $Q = 135 \text{ kJ}$

Q12.

Given: $Q = 12\,000 \text{ J}$ supplied to each; $m = 0.5 \text{ kg}$; $c_{\text{water}} = 4200$, $c_{\text{iron}} = 480 \text{ (J kg}^{-1} \text{ K}^{-1})$.

Required: the temperature rise $\Delta\theta$ of each.

Formula: $Q = mc\Delta\theta \Rightarrow \Delta\theta = \frac{Q}{mc}$.

Working:

$$\begin{aligned}\Delta\theta_{\text{water}} &= \frac{12000}{0.5 \times 4200} = \frac{12000}{2100} \approx 5.7 \text{ K} \\ \Delta\theta_{\text{iron}} &= \frac{12000}{0.5 \times 480} = \frac{12000}{240} = 50 \text{ K}.\end{aligned}$$

The iron warms about $50/5.7 \approx 8.75$ times as much, because its specific heat capacity is much smaller: the same heat causes a far bigger temperature rise in iron than in water.

Answer: $\Delta\theta_{\text{water}} \approx 5.7 \text{ K}$, $\Delta\theta_{\text{iron}} = 50 \text{ K}$

Part C — Method of mixtures

Q13.

Given: hot water $m_1 = 0.4 \text{ kg}$ at 70°C ; cold water $m_2 = 0.6 \text{ kg}$ at 20°C ; common c (water).

Required: the final temperature θ .

Formula: $Q_{\text{lost}} = Q_{\text{gained}}$: $m_1c(70 - \theta) = m_2c(\theta - 20)$ (c cancels).

Working:

$$\begin{aligned}0.4(70 - \theta) &= 0.6(\theta - 20) \\ 28 - 0.4\theta &= 0.6\theta - 12 \\ 40 &= 1.0\theta \\ \theta &= 40^\circ\text{C}.\end{aligned}$$

This lies between 20 and 70°C , as expected.

Answer: $\theta = 40^\circ\text{C}$

Q14.

Given: copper $m_c = 0.5 \text{ kg}$, $c_c = 390 \text{ J kg}^{-1} \text{ K}^{-1}$ at 95°C ; water $m_w = 0.3 \text{ kg}$, $c_w = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ at 22°C ; calorimeter of negligible heat capacity.

Required: the final temperature θ .

Formula: $Q_{\text{lost by copper}} = Q_{\text{gained by water}}$: $m_cc_c(95 - \theta) = m_wc_w(\theta - 22)$.

Working:

$$\begin{aligned}0.5 \times 390 \times (95 - \theta) &= 0.3 \times 4200 \times (\theta - 22) \\ 195(95 - \theta) &= 1260(\theta - 22) \\ 18525 - 195\theta &= 1260\theta - 27720 \\ 46245 &= 1455\theta \\ \theta &\approx 31.8^\circ\text{C}.\end{aligned}$$

This lies between 22 and 95°C , as expected.

Answer: $\theta \approx 31.8^\circ\text{C}$

Q15.

Given: aluminium cup $m_{\text{cup}} = 0.15 \text{ kg}$, $c_{\text{Al}} = 900$; water $m_w = 0.25 \text{ kg}$, $c_w = 4200$ (both all at 18°C); iron block $m_{\text{Fe}} = 0.2 \text{ kg}$, $c_{\text{Fe}} = 480$ at 150°C ; no heat lost. (c in $\text{J kg}^{-1} \text{ K}^{-1}$.)

Required: the common temperature θ .

Formula: $Q_{\text{lost by iron}} = Q_{\text{gained by cup}} + Q_{\text{gained by water}}$: $m_{\text{Fe}}c_{\text{Fe}}(150 - \theta) = (m_{\text{cup}}c_{\text{Al}} + m_wc_w)(\theta - 18)$.

Working:

$$\begin{aligned}0.2 \times 480 \times (150 - \theta) &= (0.15 \times 900 + 0.25 \times 4200)(\theta - 18) \\96(150 - \theta) &= (135 + 1050)(\theta - 18) \\96(150 - \theta) &= 1185(\theta - 18) \\14400 - 96\theta &= 1185\theta - 21330 \\35730 &= 1281\theta \\\theta &\approx 27.9^\circ\text{C}.\end{aligned}$$

This lies between 18 and 150°C , as expected.

Answer: $\theta \approx 27.9^\circ\text{C}$

Part D — Latent heat

Q16.

Given: $m = 0.25\text{ kg}$ ice at 0°C ; $L_f = 3.34 \times 10^5\text{ J kg}^{-1}$.

Required: the heat needed to melt it Q .

Formula: at the melting point $Q = mL_f$.

Working:

$$\begin{aligned}Q = mL_f &= 0.25 \times 3.34 \times 10^5 \\&= 83\,500\text{ J} = 83.5\text{ kJ}.\end{aligned}$$

Answer: $Q = 83.5\text{ kJ}$

Q17.

Given: $m = 0.2\text{ kg}$ ice at 0°C to water at 30°C ; $L_f = 3.34 \times 10^5\text{ J kg}^{-1}$, $c_w = 4200\text{ J kg}^{-1}\text{ K}^{-1}$.

Required: the total heat Q .

Formula: melt the ice ($Q_1 = mL_f$), then warm the melt-water ($Q_2 = mc_w\Delta\theta$); $Q = Q_1 + Q_2$.

Working:

$$\begin{aligned}Q_1 = mL_f &= 0.2 \times 3.34 \times 10^5 = 66\,800\text{ J} \\Q_2 = mc_w\Delta\theta &= 0.2 \times 4200 \times 30 = 25\,200\text{ J} \\Q = Q_1 + Q_2 &= 92\,000\text{ J} = 92\text{ kJ}.\end{aligned}$$

Answer: $Q = 92\text{ kJ}$

Q18.

Given: $m = 0.5\text{ kg}$ water already at 0°C ; $L_f = 3.34 \times 10^5\text{ J kg}^{-1}$.

Required: the heat that must be removed Q .

Formula: freezing removes the latent heat of fusion, $Q = mL_f$.

Working:

$$\begin{aligned}Q = mL_f &= 0.5 \times 3.34 \times 10^5 \\&= 167\,000\text{ J} = 167\text{ kJ removed}.\end{aligned}$$

Answer: $Q = 167\text{ kJ removed}$

Q19.

Given: power $P = 500\text{ W}$; time $t = 2\text{ min} = 120\text{ s}$; $L_f = 3.34 \times 10^5\text{ J kg}^{-1}$; no losses.

Required: the mass of ice melted m .

Formula: heat supplied $Q = Pt$; this melts the ice, $Q = mL_f$, so $m = \frac{Q}{L_f}$.

Working:

$$\begin{aligned}Q &= Pt = 500 \times (2 \times 60) = 60\,000 \text{ J} \\m &= \frac{Q}{L_f} = \frac{60\,000}{3.34 \times 10^5} \\&\approx 0.180 \text{ kg} = 180 \text{ g}.\end{aligned}$$

Answer: $m \approx 180 \text{ g}$

Q20.

Given: $m = 0.3 \text{ kg}$ water at 50°C to steam at 100°C ; $c_w = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$.

Required: the total heat Q .

Formula: warm the water ($Q_1 = mc_w \Delta\theta$), then boil it ($Q_2 = mL_v$); $Q = Q_1 + Q_2$.

Working:

$$\begin{aligned}Q_1 &= mc_w \Delta\theta = 0.3 \times 4200 \times 50 = 63\,000 \text{ J} \\Q_2 &= mL_v = 0.3 \times 2.26 \times 10^6 = 678\,000 \text{ J} \\Q &= Q_1 + Q_2 = 741\,000 \text{ J} = 741 \text{ kJ}.\end{aligned}$$

Answer: $Q = 741 \text{ kJ}$

Q21.

Given: $m = 0.6 \text{ kg}$ water at 0°C frozen to ice at 0°C ; time $t = 4 \text{ min} = 240 \text{ s}$; $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$.

Required: the average rate of heat removal (W).

Formula: heat removed $Q = mL_f$; rate $= \frac{Q}{t}$.

Working:

$$\begin{aligned}Q &= mL_f = 0.6 \times 3.34 \times 10^5 = 200\,400 \text{ J} \\t &= 4 \times 60 = 240 \text{ s} \\\text{rate} &= \frac{Q}{t} = \frac{200\,400}{240} \approx 835 \text{ W}.\end{aligned}$$

Answer: $\text{rate} \approx 835 \text{ W}$

Part E — Graph reading

Q22.

Given: $m = 0.5 \text{ kg}$ pure solid; segment AB warms it from 0 to 50°C absorbing $Q_{AB} = 21\,000 \text{ J}$; plateau BC at 50°C ; plateau DE at 120°C absorbing $Q_{DE} = 565\,000 \text{ J}$.

Required: (a) the melting and boiling points; (b) the specific heat capacity c of the solid; (c) the specific latent heat of vapourisation L_v .

(a) The flat parts mark the changes of state, so the **melting point** is 50°C (plateau BC) and the **boiling point** is 120°C (plateau DE).

Answer: $\text{melting point } 50^\circ\text{C}, \text{ boiling point } 120^\circ\text{C}$

(b)

Formula: segment AB warms the solid, so $Q_{AB} = mc \Delta\theta \Rightarrow c = \frac{Q_{AB}}{m \Delta\theta}$ with $\Delta\theta = 50 - 0 = 50 \text{ K}$.

Working:

$$\begin{aligned}c &= \frac{Q}{m \Delta\theta} = \frac{21\,000}{0.5 \times 50} \\&= \frac{21\,000}{25} = 840 \text{ J kg}^{-1} \text{ K}^{-1}.\end{aligned}$$

Answer: $c = 840 \text{ J kg}^{-1} \text{ K}^{-1}$

(c)

Formula: plateau DE is the boiling (vapourisation) stage, so $Q_{DE} = mL_v \Rightarrow L_v = \frac{Q_{DE}}{m}$.

Working:

$$\begin{aligned} L_v &= \frac{Q}{m} = \frac{565000}{0.5} \\ &= 1.13 \times 10^6 \text{ J kg}^{-1}. \end{aligned}$$

Answer: $L_v = 1.13 \times 10^6 \text{ J kg}^{-1}$